

NITIN SHYAMKUMAR

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Objective: Obtain software engineering position in high growth organization

EDUCATION

2020 - 2022	NEW YORK UNIVERSITY, Courant Institute of Mathematical Sciences M.PHIL., Computer Science (GPA: 3.95) AWARDS: MacCracken Fellowship (Fully funded Ph.D. candidate)
2013 - 2017	CORNELL UNIVERSITY, College of Arts and Sciences B.A. Mathematics, B.A. Computer Science (GPA: 4.01) AWARDS: Phi Beta Kappa, Tanner Dean's Scholar

EXPERIENCE

2021	Sensorum Technologies SOFTWARE ENGINEER INTERN - Responsible for HIPAA-compliant product MVP's backend at Series A startup - Built data aggregation AWS Lambda endpoints, test suite infrastructure, and custom deployment procedures via Github Actions (NodeJS, Postgres)
2017 - 2020	Applied Predictive Technologies SENIOR SOFTWARE ENGINEER - Team colead for statistical modeling tools and pipelines (C#, SQL, R, C++, React, Typescript) - Led end to end feature development for internal modeling pipeline API with 4 FTE (C#, SQL) - Prototyped 4x speedup in linear regression calculations (C++, C#, Fortran) - Co-led research into deep learning for uplift measurement on financial transaction data (Keras, C#, R) - Co-led Cornell recruiting team, and team management

PUBLICATIONS

"Towards context-aware learning for control: Balancing stability and model-learning error" Shyamkumar, Nitin , Serkan Gugercin, and Benjamin Peherstorfer 2022 <i>American Control Conference</i> . IEEE, 2022
"Automated home cage training of mice in a hold-still center-out reach task." Bollu, Tejaspratap, Samuel C Whitehead, Nikil Prasad, Jackson Walker, Nitin Shyamkumar , Raghav Subramaniam, Brian Kardon, Itai Cohen, Jesse H Goldberg <i>Journal of Neurophysiology</i> . 121(2), 500-512, 2019
"Sublinear estimation of a single element in sparse linear systems" Shyamkumar, Nitin , Siddhartha Banerjee, and Peter Lofgren 2016 <i>54th Annual Allerton Conference on Communication, Control, and Computing (Allerton)</i> . IEEE, 2016

PROJECTS

Low rank neural network architectures for fast solutions to parametric partial differential equations Shyamkumar, Nitin , Yuxiao Wen, and Benjamin Peherstorfer
Solving control problems using physics informed machine learning for partial differential equations Shyamkumar, Nitin , Arasu Arun, and Leo Wu
Steady state analysis of BBR using the Network Calculus framework Shyamkumar, Nitin

SKILLS AND INTERESTS

Skills: Research • Full stack software engineering • Modeling and data pipelines • Numerical linear algebra • GPU programming • Deep learning • Causal inference
Languages: C# • Python • Javascript • Typescript • NodeJS • C++ • R
Frameworks: Pytorch • Jax • CUDA • ReactJS • Redux • Redux-Saga • AWS Lambda
Interests: Brazilian jiu jitsu • Long distance running without shin splints • Perfectly pan frying salmon